DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #04E: WEAPONS ARSENAL (PART 5 — HOLISTIC WEAPON ECOSYSTEM, SOUL-FORGE SPIRITS, KILL-EVOLUTION, RUNE SOCKETING, 51 RELICS & CATCH-ALL AUDIT)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Series Completion (`04A–04E` Complete 16-Family + Holistic Weapon Ecosystem Audit)**:
  - `04A`: Group 1 — `SWORD`, `BLADE`, `SPEAR`, `STAFF`
  - `04B`: Group 2 — `FIST`, `CLAW`, `BOW`, `NEEDLE`, `WHIP`
  - `04C`: Group 3 — `FOCUS`, `ZITHER`, `BRUSH`, `ROSARY` + Crafting Material Ecology
  - `04D`: Group 4 — `FAN`, `ASTROLABE`, `ABACUS` + 7 Inert `v2.0.0` Debuffs
  - **`04E` (This Final Weapon Document)**: **Holistic Cross-System & Catch-All Weapon Audit** — covering everything shared across or outside the 16 families: Instant Armory Fire-Sale Deletion, Kill-Farming Evolution Exploit, Dual Legendary Relic Universes (`20 VC` vs `31 REL_V230_*`), Sentient Weapon Spirits (`SECT.SOUL_FORGE` & `ARMAMENT_WILLS`), Elemental Rune Socketing, Regex Classification & Hybrid Implements (`Hammer`, `Scythe`, `Lantern`, `Banner`), Magnetic Disarm Physics, Enemy Weapon Parity, Remote Repair & Save/Load Swap Determinism.

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order & Closed-Loop Conservation**: Preserve `<script>` order in `index.html`; never destroy rare manuals/relics as generic crafting fodder (`socketElementalRunes`), and never delete newly acquired armory weapons via stale-stock miscalculation.
3. **100% Emergent NPC & Artifact Agency**: Weapons in a living Xianxia world have provenance, physical laws, durability logistics, and (at high tiers) sentient wills (器灵) that remember their fallen masters and judge new wielders!

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1. [WPN-SYS-01] Critical Armory Deletion Bug: EXT.sellStaleSectGear Liquidates Brand-New Bought, Researched & Fished Weapons in the Same Millisecond!

- **Locations**: `js/economy.js:244`, `js/external_market.js` (`EXT.tradeSect` & `EXT.sellStaleSectGear`), `js/content.js`, `js/fishing.js`
- **Root Cause**:
  - `ECO.finishCraft` (`economy.js:285`) and `ECO.craftProceduralNow` (`line 781`) manually set `gear.listedAtDay = W.day`, **but `ECO.mintGear` (`line 237`) and `ECO.rollProceduralGear` (`line 715`) do NOT set `listedAtDay`!**
  - Consequently, whenever the sect buys a weapon from a merchant (`EXT.tradeSect`), discovers a prototype weapon via `CONTENT` research (`content.js`), or fishes up a chest weapon (`fishing.js`), `gear.listedAtDay` is `undefined` (`0`).
  - At the bottom of `EXT.tradeSect`, `EXT.sellStaleSectGear` checks `W.day - (it.listedAtDay || 0) >= 6` (`W.day >= 6` is true for 99% of the game!) and **immediately sells the brand-new weapon back to the exact same merchant at a `-35%` loss and deletes it from the world**!
  - **Even Worse Risk on Heirlooms**: Any weapon with `imbuedElderSoul` (`PROP-31`), `weaponSpirit` (`SOUL_FORGE`), or `REL_V230_*` sitting in `W.sect.gear` while disciples are away risks liquidation unless explicitly protected!
- **Exact Fix**:
  1. Stamp `listedAtDay: W ? W.day : 0` directly inside **both** `ECO.mintGear` and `ECO.rollProceduralGear` (and whenever pushing into `W.sect.gear`).
  2. Hard-exempt `it.legendary || it.unique || it.isLegendaryRelic || it.weaponSpirit || it.imbuedElderSoul || it.isSectHeritage || it.soulBoundOwnerId` from `EXT.sellStaleSectGear` permanently!

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2. [WPN-SYS-02] Unfixed Universal Kill-Evolution Exploit (EQ-02 in ECO.recordWeaponKill, js/economy.js:1079–1097)

Inspecting live `js/economy.js:1079–1097` confirms that **every ordinary and procedural weapon in the game evolves up to +3 Full Tiers (`Tier 4 -> Tier 7 Mythic!`) simply by killing 25 Tier-1 wild animals**:

```javascript
const nextStage = wpn.killTracker.kills >= 25 ? 3 : (wpn.killTracker.kills >= 10 ? 2 : (wpn.killTracker.kills >= 3 ? 1 : 0));
if (nextStage > wpn.killTracker.evolutionStage) {
  wpn.killTracker.evolutionStage = nextStage;
  wpn.tier = Math.min(7, (wpn.tier || 1) + 1);
  wpn.q = Math.round(((wpn.q || 1) + 0.08) * 100) / 100;
  ...
```

- **Four Severe Progression & Logic Bugs**:
  1. **Destroys the Entire T5–T7 Crafting & Rare-Ore Economy**: `procedural` gear is designed to cap at **Tier 4** so T5–T7 weapons require `MAT_ORE_METEOR_IRON_5`, `MAT_ORE_DIVINE_GOLD_6`, `MAT_ORE_PRIMORDIAL_CRYSTAL_7`, and high `FORGING` skill. Farming 25 Tier-1 boars turns a cheap Tier-4 iron blade into a **Tier-7 weapon** for zero materials!
  2. **Zero Victim-Strength Gate**: Slaying a Tier-1 rabbit (`victim.tier === 1`) counts toward evolving a Tier-6 weapon to Tier-7!
  3. **`minStage` & `combatMult` Desynchronization**: `wpn.tier` increases up to `+3`, **but `wpn.minStage` stays at its original low value** (allowing a Stage-1/2 novice to keep wielding an evolved high-tier weapon!) and `wpn.combatMult` (`economy.js:709`) is never recomputed when `delete wpn.st` runs!
  4. **Mutates Canonical Tier-6 Legendaries**: Unique canonical Tier-6 relics (`FULL_MOON_SCIMITAR`, `DAGOU_JADE_STAFF`, `OVERLORD_SPEAR`, `WRATHFUL_TIGER_GAUNTLETS`, `CONDOR_METEOR_BOW`) lack `noEvolution: true`, mutating into Tier-7 `q = 1.54` after 25 trivial kills!
- **Balanced Wuxia Solution**:
  - **Ordinary / Procedural Weapons (`!wpn.livingWeapon && !wpn.weaponSpirit`)**: Cap veteran battle-tempering (ลับคมด้วยโลหิตศึก) to **quality honing only (`+0.04 q` per threshold, max `+0.12 q`, `0` tier jump)**, preserving the Tier-4 procedural ceiling and T5–T7 forging economy!
  - **True Living / Soul-Forged / Blood-Tempered Weapons (`wpn.livingWeapon || wpn.weaponSpirit || wpn.imbuedElderSoul`)**: Can evolve Tier (`max`

`+2 tiers`), **requiring worthy prey (`victimTier >= wpn.tier - 1`, and `bossKills >= 1` for Tier 6 $\to$ 7)** while syncing `combatMult` and `minStage`!

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3. `[WPN-SYS-03]` The Two Disjoint Legendary Weapon Universes (`20 VC.LEGENDARY_WEAPONS` vs `31 REL_V230_<PATH>` = `51` Relics!)

`JS2-v2.6.0` currently contains **two separate Legendary Weapon architectures** that do not obey the same world laws:

Property / Rule	**`20 VC.LEGENDARY_WEAPONS` (`visitors_clues.js`)**	**`31 REL_V230_<PATH>` Path Relics (`data_martial.js`)**
World Uniqueness (`uniqueArtifacts`)	Strictly **1 copy in the entire world** (`unique: true`)	**Repeatable Forge Recipes (`REC_REL_V230_*`)** — can mass-forge duplicates!
Clue Quest Discovery (`CLUE_TEMPLATES`)	`14/20` have quests (**`6` `v2.0.0` relics have `0` quests!**)	**`0/31` have clue quests**
Combat Proc Engine	`VC.runWeaponProcs` (`visitors_clues.js`)	Ignored by `VC.runWeaponProcs`; relies on `grantedSkills: ['LEG_V230_<PATH>']`
3/3 Primordial Trinity (`evaluateLegendaryReliCResonance`)	Counts (`it.legendary === true`)	**Also counts (`it.legendary === true`)**, making 3/3 Trinity mass-producible!
12 Founding Paths Coverage	Covered by Wuxia literary relics (`Yitian`, `Tulong`, etc.)	**None** (`REL_V230_*` only exists for paths `13..43`)

- **Remediation**:
  1. Mark `REC_REL_V230_<PATH>` as **Sect-Unique Grandmaster Forgings** (if an active non-`playerSpawned` instance of `REL_V230_<PATH>` already exists in `W.world.uniqueArtifacts` / sect, block duplicate forging in

`ECO.craftEligible` and register `REL_V230_<PATH>` in `VC.syncUniqueArtifacts`!).
2. Author the **6 missing Clue Chains** in `VC.CLUE_TEMPLATES` (`ROSARY`, `CLAW`, `NEEDLE`, `FAN`, `ASTROLABE`, `ABACUS` — detailed in `04B–04D`) so all `20/20` `VC.LEGENDARY_WEAPONS` are discoverable via living Jianghu rumors!

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4. `[WPN-SYS-04]` Sentient Weapon Spirits & Ancestral Soul Imbuement (`SECT.SOUL_FORGE` & `ARMAMENT_WILLS`)

- **Locations**: `js/sect.js` (`SECT.SOUL_FORGE`), `js/ai.js` (`AI.kill`), `js/visitors_clues.js:1075–1086`
- **Three Structural & Contextual Gaps**:
    1. **Zero Autonomous Spirit Awakening (`awaken-weapon-soul` Is Player-UI Only!)**:
        - Neither `SPIRIT_SWORD_ARTIFICER`, `PUPPET_RELIC_ARTIFICER`, `forge` Elders, nor Stage-6+ Grandmasters ever autonomously call `SECT.SOUL_FORGE.awakenWeaponSpirit` on their cherished T5–T7 weapons! Add an autonomous goal/forge rite when an Elder (`stage >= 5`) has fought ≥ 10 battles with a T5+ weapon.
    2. **Death-Strip Race Condition (`AI.kill` / `VC.syncUniqueArtifacts` vs `imbueElderSoulIntoWeapon`)**:
        - When an Elder dies, if `VC.syncUniqueArtifacts` (`visitors_clues.js:1076–1082`) or loot cleanup strips `n.equip` into `W.sect.gear` (`n.equip = {}`) **before** `imbueElderSoulIntoWeapon(W, n, n.equip.WEAPON)` executes, the dying Elder's equipped weapon reference is lost! Always run `imbueElderSoulIntoWeapon` & `inheritSentientWeaponOnDeath` **first inside `AI.kill` before stripping `n.equip`**, and prefer bequeathing the sentient heirloom directly to the Elder's **personal `apprentice` / `RED_THREAD` spouse / bloodline child** before dumping it into generic storage!
    3. **Sentient Weapon Alignment & Grudge Rejection (器灵择主 – ศาสตราเลือกนาย)**:
        - Currently, if a righteous `BUDDHA_DHARMA` Elder's soul resides inside a weapon (`imbuedElderSoul`), even a corrupt Demonic traitor—or **the rival who murdered that Elder in a `grudge` duel**—can wield it at `100%` power!
        - Wire `evaluateSoulBoundCondition` (`economy.js:1126`): if a weapon holds `imbuedElderSoul` (or `weaponSpirit`), grant **+12% resonance** to the Elder's disciples/kin/aligned-integrity heirs, and inflict

Spirit Rejection (`powerMult: 0.65`, `heartDemon` backlash) on anyone holding a `BLACK_THREAD` grudge against the imbued soul or extreme moral opposition!

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5. `[WPN-SYS-05]` Elemental Rune Socketing (`ECO.socketElementalRunes`, `js/economy.js:1244–1311`): Destroys Priceless Items & Zero AI Usage!

- **Exact Code Bug (`economy.js:1265–1269`)**:

```javascript
const matKeys = Object.keys(W.sect.stock || {}).filter(k => k !== 'grain' && k !== 'meal' && !ECO.isPill(k) && (W.sect.stock[k] || 0) > 0);
if (!ECO.take(W, matKeys[0], 1)) return { ok: false, reason: 'insufficient-resources' };
```

- **Bug 1 (Destroys Path Manuals & Tier-7 Relics!)**: `matKeys[0]` grabs **the first arbitrary key in `W.sect.stock`** that isn't grain/meal/pill—so engraving a 30-stone rune can literally consume **`MANUAL_PATH_CHRONO_EPOCH`, `CRYSTAL_PATH_VOID_SPACE`, or `MAT_TR_WORLD_FORGE_SHARD` (`5,200` stones)**! Restrict rune material consumption strictly to **Tier 2–4 Elemental Ores / Formation Parts (`category === 'ORE' || category === 'FORMATION_PART'`, `tier <= 4`, `!m.relic`, above `ECO.reserveFloor`)** matching at least one socketed element!
- **Bug 2 (`'DARK_YIN'` vs `'YIN_SHADOW'` Rejection)**: Normalize `'YIN_SHADOW'` $\leftrightarrow$ `'DARK_YIN'` before calling `CE.resolveElementalReactionSpec` so Yin-element runes don't fail with `'no-valid-reaction'`!
- **Bug 3 (Zero NPC Autonomy)**: Wire `AI.ACTS.engraveWeaponRunes` at `forge` so `RUNE_ARRAY_ENGRAVER` / `forge` masters (`FORGING >= 45` & `ARRAY/TALISMAN >= 35`) socket matching 2-element runes onto un-socketed T4+ weapons!

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6. [WPN-SYS-06] Catch-All Classification, Hybrid Weapons, Magnetic Disarm Physics & Save/Load Hygiene

2. **Harden `JD.MARTIAL.classify(slot, nameTh)` Against Cross-Slot & Uncategorized Collisions**:

    - Always evaluate `if (slot === 'ARMOR') ...` and `if (slot === 'ACCESSORY') ...` **before** any weapon regex so `"ประคำ"`, `"จาน"`, `"ขลุ่ย"`, and `"ตรา"` on Accessories never leak into `ROSARY` / `ASTROLABE` / `ZITHER` / `FOCUS`!
    - Explicitly classify hybrid/non-standard weapon nouns inside `slot === 'WEAPON'`:
        - `/ค้อน|กระบอง|ลูกตุ้ม|คฑาหนัก/` $\to$ `'STAFF'` (`visualSubtype: 'HAMMER'`, `HEAVY_BREAKER` guard-shatter, instead of falling through to `'SWORD'`!)
        - `/เคียวโซ่/` $\to$ `'WHIP'`; `/เคียว|ขวาน/` $\to$ `'BLADE'` (`HEAVY_BREAKER`)
        - `/กริช|มีดสั้น|มีดบิน/` $\to$ `'BLADE'` (`visualSubtype: 'DAGGER'`)
        - `/ขลุ่ย|สังข์|ระฆังศึก|กลองศึก/` $\to$ `'ZITHER'`
        - `/โคม|แส้ปัด|กระจก|ลูกแก้ว/` $\to$ `'FOCUS'`
        - `/ธงศึก|ธงรบ/` $\to$ `'SPEAR'`

3. **Magnetic Weapon Disarm Physics (`EFX_WEAPON_MAGNET_DISARM` on `ITEM190_TAL_010` & `ITEM190_CORE_007`)**:

    - Currently either unapplied in combat or indiscriminate. Wire `EFX_WEAPON_MAGNET_DISARM` so magnetic fields (`EARTH_METAL` + `LIGHTNING`) strongly suppress **metallic weapons (`SWORD`, `BLADE`, `SPEAR`, `CLAW`, `NEEDLE`, `ABACUS` — `-35%` weapon `atk` for 6 ticks)**, have **zero effect** on non-metallic wood/jade/silk implements (`ZITHER`, `BRUSH`, `ROSARY`, wooden `STAFF`) or wild beasts (`u.wild === true`), and give Rival/Foe cultivators explicit `foe.weaponType` tags so weapon tactics & disarm interactions work symmetrically against enemies!

4. **Block Cross-Continent Remote Repair (`sim.js`) & Persist `lastGearSwapAt` (`N-08`)**:

    - In `SIM.hourly`, require `!n.away` before pulling `ITEM_REPAIR_STONE_*` from `W.sect.stock` (away NPCs may only use repair stones in their own `n.inv.mats`!).

- Rename transient `_lastGearSwapAt` to persisted `lastGearSwapAt` so save/load roundtrips remain 100% deterministic without post-load weapon-swapping spikes!

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7. Master Summary of the Complete `04A–04E` Weapons Arsenal Audit Series

Document	Weapon Families / Systems Audited	Key Highlights
`04A`	`SWORD`, `BLADE`, `SPEAR`, `STAFF`	`XUANTIE` heavy-sword tactics, `XIAOLI` zero-Qi Samurai lockout & melee range bug, `DAGOU` 3-armor copy-paste bug, non-procedural `dur=1` (`30%` stat rot)
`04B`	`FIST`, `CLAW`, `BOW`, `NEEDLE`, `WHIP`	`FIST`/`CLAW` name collision, `GOLDEN_NEEDLE_SAINT` rejecting Needles (`0.72×`), `CONDOR_BOW` trial missing `BOW` skill, `TAOTIE`/`BAIHUA` missing clues, `WHIP` stale IDs
`04C`	`FOCUS`, `ZITHER`, `BRUSH`, `ROSARY`	Mining weasel fur & farming python tendons (`-35%` herb starvation!), `CHUNQIU_BRUSH` `armorbreak` vs `mdefbreak` bug, `KARMA_SAMSARA` rejecting `ROSARY`
`04D`	`FAN`, `ASTROLABE`, `ABACUS`	`tempoBreak`, `contractDefault`, & `gravityCrush` 100% inert in `stats.js`/`combat.js`, raw Hanzi in `BAJIAO_FAN`, missing clues & economic/audit utility

Document	Weapon Families / Systems Audited	Key Highlights
04E	**Holistic Ecosystem & Catch-All**	`sellStaleSectGear` 0-ms deletion, 25-rabbit T7 evolution exploit, `20 VC` vs `31 REL_V230_*` split, `SOUL_FORGE` death race & will rejection, rune stock-destruction